

UPPER STEWIACKE, NS, Canada

WMO: 717530

Lat: 45.2326N

Lon: 63.0546W

Elev: 77

StdP: 14.66

Time Zone: -4.00 (NAA)

Period: 06-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-5.9	-0.4	-11.7	2.9	-3.0	-6.8	3.8	2.6	27.3	34.3	24.3	35.1	2.3	40	0.587

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.0	83.1	68.7	80.4	67.4	77.9	66.1	71.8	79.5	70.2	76.8	68.5	74.4	8.9	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
69.2	107.6	75.8	67.7	102.2	73.8	66.2	96.9	72.2	35.6	79.6	34.2	77.1	32.8	74.5	81.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.9	18.8	16.4	-15.4	88.0	4.9	1.9	-18.9	89.3	-21.8	90.4	-24.6	91.5	-28.2	92.8		
(5)			-15.8	75.4	4.9	2.1	-19.3	76.9	-22.2	78.2	-24.9	79.3	-28.5	80.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	43.9	23.0	22.4	28.9	39.7	49.7	58.2	66.2	65.0	57.4	47.3	38.4	29.0		
	DBStd	17.16	10.92	10.31	8.94	7.01	6.18	6.19	4.82	5.02	7.09	7.58	8.89	10.41		
	HDD50	3851	838	773	654	319	82	7	0	0	14	146	365	653		
	HDD65	7880	1303	1193	1118	759	476	217	45	62	242	550	798	1117		
	CDD50	1619	0	0	1	10	72	254	501	465	237	62	16	1		
	CDD65	173	0	0	0	0	1	14	81	61	15	1	0	0		
	CDH74	1583	0	0	1	4	37	148	725	552	114	2	0	0		
	CDH80	265	0	0	0	0	6	17	138	89	15	0	0	0		
	WSAvg	6.7	7.3	7.5	8.3	8.0	7.2	6.3	5.4	5.0	5.5	6.2	6.9	7.1		
	PrecAvg	51.6	4.7	3.8	4.1	3.7	3.7	3.9	4.3	3.6	4.5	5.2	5.6	5.6		
(15) Precipitation	PrecMax	63.4	9.1	6.7	6.6	6.7	6.7	10.6	7.7	13.2	6.0	9.1	10.1	9.8		
	PrecMin	36.1	1.2	1.2	1.1	0.7	1.3	1.0	0.8	0.6	0.9	1.2	1.3	3.0		
	PrecStd	8.1	1.7	1.5	1.5	1.5	1.6	1.8	2.0	2.4	1.5	2.0	2.0	1.9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.5	52.4	58.5	68.0	79.1	82.6	87.0	86.0	82.8	72.7	64.3	57.9		
		MCWB	53.1	50.3	49.0	55.3	61.5	65.0	71.1	71.2	69.9	63.6	60.8	55.5		
	2%	DB	48.4	45.6	51.0	61.5	71.9	78.0	83.6	82.4	77.4	68.0	61.3	53.1		
		MCWB	47.0	43.1	45.5	52.4	59.0	64.9	69.6	68.7	66.6	61.2	58.7	51.4		
	5%	DB	43.4	40.9	46.1	57.1	66.7	74.8	81.0	79.7	73.8	64.4	58.3	48.4		
		MCWB	41.4	37.8	41.9	49.7	55.3	62.9	67.8	67.2	64.9	59.8	55.8	46.8		
	10%	DB	38.1	36.6	42.0	53.2	62.7	71.3	78.2	77.0	70.7	61.3	54.5	43.7		
		MCWB	36.2	34.5	37.7	47.3	53.9	61.7	66.8	65.8	64.2	56.7	51.5	41.7		
	0.4%	WB	53.3	50.6	51.8	59.3	65.4	71.0	73.8	73.6	71.9	67.4	61.7	56.1		
		MCDB	54.5	52.1	54.5	65.5	74.9	78.4	82.6	81.5	80.0	69.4	63.4	57.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	47.4	44.0	46.7	55.3	61.3	68.0	72.1	71.5	69.4	63.1	59.2	51.3		
		MCDB	48.3	45.3	49.2	59.1	67.5	73.5	80.0	78.7	73.6	66.0	61.1	52.9		
	5%	WB	41.8	38.4	42.3	51.6	58.0	65.5	70.4	69.7	67.6	60.5	56.5	46.8		
		MCDB	43.2	40.3	45.5	55.6	63.5	71.5	77.6	76.8	71.6	63.4	58.2	48.1		
	10%	WB	36.5	34.7	38.1	48.1	55.4	63.0	68.7	67.7	65.4	57.2	51.9	42.3		
		MCDB	37.9	36.4	41.6	51.6	61.0	69.2	75.0	74.4	69.6	60.9	53.9	43.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	18.4	18.6	18.7	19.7	21.5	21.2	22.0	22.6	23.0	20.3	17.0	15.9		
		MCDBR	22.6	20.2	24.4	28.2	30.6	28.5	27.6	26.6	26.1	23.1	19.9	20.6		
		MCWBR	22.5	18.0	19.2	19.3	18.2	16.6	15.0	14.9	16.4	16.9	17.5	20.3		
		MCDBR	22.1	18.9	22.7	22.5	22.8	21.2	21.6	22.4	19.3	19.5	17.7	20.4		
	5% WB	MCWBR	22.2	18.0	19.8	18.0	15.6	13.4	12.9	13.5	13.7	16.1	16.9	20.4		
(40) Clear-Sky Solar Irradiance	taub		0.298	0.293	0.322	0.353	0.369	0.397	0.414	0.397	0.351	0.331	0.314	0.291		
	taud		2.451	2.484	2.448	2.411	2.410	2.356	2.330	2.375	2.481	2.532	2.530	2.525		
	Ebn at Noon		259	285	290	288	285	276	270	271	276	267	252	252		
	Edh at Noon		22	26	31	35	36	38	39	36	30	24	21	19		
(44) All-Sky Solar Radiation	RadAvg		435	697	1079	1326	1546	1679	1698	1562	1239	793	446	328		
	RadStd		37	58	71	118	182	153	121	115	96	84	49	23		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (37 neighbors)		N/A	N/A	N/A	N/A	N/A	+0.79	N/A	N/A	+97	+29	

Nomenclature: See separate page